B.Voc. (Food Science and Technology)
Session 2016-17

Semester I

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN - 101	Communication Skills	Generic	Theory	20	80	--	--	6
*GEN-102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	6
FST 103	Introduction To Food Microbiology	Skill	Theory & Practical	10	40	10	40	6
FST 104	Bakery And Confectionery Technology-I	Skill	Theory & Practical	10	40	10	40	6
FST 105	Bakery And Confectionery Technology-II	Skill	Theory & Practical	10	40	10	40	6

Semester II

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN -201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
FST 107	Management of Food Industry	Generic	Theory & Practical	10	40	10	40	6
FST 108	Dairy Technology	Skill	Theory & Practical	10	40	10	40	6
FST 109	Milk And Dairy Food Products Development	Skill	Theory & Practical	10	40	10	40	6
FST 110	Food Packaging	Skill	Theory & Practical	10	40	10	40	6
	Industrial Visit/ Training and Report Making							NC**

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

**** Non Credit**

FST 103

INTRODUCTION TO FOOD MICROBIOLOGY

Course Objectives: To understand the types of food microbes, causes for food spoilage process for food spoilage, need for food preservation.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Introduction to Food Microbiology

- Definition and Scope of food microbiology
- Characteristics of Microorganisms in Food
- Types of microorganisms, Classification and Nomenclature, Morphology and Structure and their importance in food (bacteria, fungi, viruses and prions, protozoans and others)
- Significance of spores
- Heat resistance of microorganisms and their spores –bacteria and their spores –fungi and their spores –yeast and their spores - spoilage of canned foods and types of spoiled cans

Unit II

- Microbial Growth in Food
- Microbial Growth Characteristics- Bacterial growth curve, microbial reproduction and microbial growth in food
- Factors affecting the growth of micro organisms in food

Unit III

Microbial Food Spoilage

- Sources of Microorganisms in foods
- Some important food spoilage bacteria
- Food spoilage: Microbial, physical, chemical & miscellaneous.
- Changes caused by micro-organisms during spoilage (breakdown of proteins, carbohydrates, fats and other constituents)
- Spoilage of specific food groups- a) Cereal and cereal products b) Vegetables and fruits c) Meat and meat and seafood products d) Milk and dairy products e) Egg and egg products f) Canned foods.
- Detection of food pathogens**: Overview of Conventional and Rapid methods to detect food pathogens.**

Unit IV

Control of Microorganisms in Foods

- Principles and methods of preservation
- Physical Methods of Food Preservation- Dehydration, Freezing, Cool Storage, Heat Treatment (esp. thermo bacteriology), Irradiation,
- Chemical Preservatives

- Bio preservatives
- New Non Thermal methods
- Introduction to Hurdle concept and Predictive Microbiology Practical:

1. Introduction to the Basic Microbiology Laboratory Practices and Equipments
2. Functioning and use of compound microscope
3. Cleaning and sterilization of glassware
4. Preparation and sterilization of nutrient broth
5. Cultivation and sub-culturing of microbes
6. Preparation of slant, stab and plates using nutrient agar
7. Morphological study of bacteria and fungi using permanent slides
8. Methods of sterilization and preparation of media
9. Simple staining
   Endospore staining
   Standard Plate Count Method

Recommended Readings

1) Frazier William C and Westhoff, Dennis C. Food Microbiology, TMH, New Delhi, 2004
2) Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000
3) Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997
4) Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 199
5) Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.
6) Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Eastern Economy Edition, 6th edition,
7) William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, The Mc Graw-Hill Companies,

FST 104
BAKERY AND CONFECTIONERY TECHNOLOGY-I

Course Objectives:

- To provide know about the machinery and process involved in the baking process
- To understand the various types of sugar and its grades
- To know the confectionery product manufacture

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

- Introduction to food processing and preservation.
- Various products of the bread and bakery sub-sector
- List the various types of industries within the bakery sub-sector
- Explain the baking process
- Equipments used in Bakery (Dough mixers, Dividers, rounders, Proofing, moulding, Ovens, Slicers etc), Cleaning and maintenance of the work area and machineries , Baking ingredients required for production and plan production sequence
- Testing of Flour For Bakery Goods: Laboratory testing of Wheat grain Quality, Moisture tests, Grain hardness testing. Testing, Visco graph, Amylograph, Ferinograph. Units of measurement used in the baking industry.
- Raw material required for bakery products. Role of flour, water, yeast, salt,Sugar milk and fats.
- Yeast- An elementary knowledge of Baker's yeast, the part it plays in th fermentation of dough and conditions influencing its working .
- Effect of over and under fermentation and under proofing of dough and other fermented goods
- Mixing methods used for baking, Calculate batch size and plan for various types of dough as per the production schedule

Unit II

- process of mixing and knead ingredients to make dough
- Oven & Baking – Knowledge and working of various types of oven.
- Biscuit-Types of biscuit dough – Developed dough, short dough, semi-sweet, enzyme modified dough and batters- importance of the consistency of the dough, Factors affecting the quality of biscuits/ cookies
- Cake – Flour specification – ingredients: Cake making ingredients- flour, sugar, shortening and egg types of chemically aerated goods Moistening agents Fats and Oil Leavening agents
- Manufacturing process – Cake making methods sugar batter process, flour batter process, genoise method and blending and rubbing method, Correct temperature for

baking different varieties of cakes, Balancing of cake formula Characteristics of cakes: external: Internal ,Types of icing

Unit III

- Types of Confectionery, raw materials and processing of chocolates, hard boiled candies.
- Additives for Confectioneries.
- Equipments used in Confectionery manufacture
- Toffees – Milk toffee; chocolate ; stick jaws; liquor chocolate.
- Pastry making, principles and various derivatives
- Baking temperatures for confectionery goods.
- Process of proofing process of baking products in the oven
- Process of cooling of baked products

Unit IV

Bread manufacturing process – Straight dough fermentation Bread improvers- improving physical quality

Methods of bread making

- Straight dough methods
- Delayed salt method
- No time dough method
- Sponge and dough method Characteristics of good bread
- External Characteristics – volume, symmetry of shape
- Internal characteristics- colour, texture , aroma, cliarity and elasticity Bread faults and their remedies.

Bread diseases- Rope and Mould – causes and prevention.

Practicals

1. Preparation of cookies and biscuits
2. Preparation of different types of cakes
3. Preparation of different types of pastries
4. Preparation of different types of ice creams (Vanilla, Strawberry, chocolate, Pineapple, Mango)
5. Preparation different types of Pudding
6. Preparation of Bread rolls; Bread sticks & softs rolls;
7. Preparation of Buns; Hot Cross Buns;
8. Preparation of Fermented doughnuts;
9. Preparation of Chelsea buns; Russian stolen Basic bun dough ,
10. Preparation of Pizza base

Reference Books

1. Samuel A. Matz, “Bakery Technology and Engineering”, Chapman & Hall, 3rd Edition, 1992.
2. Reference Books 1. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
3. Pomeranz Y, “Modern Cereal science and Technology” MVCH Publications, NY, 1987.
4. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.
5. Samuel A. Matz, “Equipment for Bakers” Pan Tech International Publication, 1988.
6. Stanley P Cauvain, Linda S Young, “Technology of Bread making”, Aspen publication, 2ndEdition, 2007.
7. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
8. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.

FST 105
BAKERY AND CONFECTIONERY TECHNOLOGY-II

nstructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

Bakery layout- The required approvals for setting up of a Bakery- Government procedure and Bye laws.

Selection of site & Selection of equipment

- Layout design
- Electricity
- Process of cleaning the work area and machineries after production
- Roles and responsibilities of a bakery production specialist.
- Plan , organize and management of work in bakery.
- Method of documenting and recording the details of baking ingredients to final finished product

Unit II

- Quality control of raw material and of finished products
- Role of Hygiene in Bakery
- Personal hygiene, care of skin, hand and feet, Food handlers hygiene, protective clothing.
- Dishwashing methods- manual and machine dish washing- merits and demerits
- Garbage disposal- different methods- advantages and disadvantages
- Food poisoning- Causative factors and the precautions to be taken by food handlers.

Unit III

- Food Storage- Techniques of correct storage, storage temperature of different commodities prevent bacterial manifestation or contamination . Storage of confectionery products.
- Pest Control- Rodents and insect control techniques, special stress on control of flies , rats and cockroaches, care of the premises and equipment
- Municipal health laws.
- Golden rules of first aid and treatment for cuts, wounds , burns

- Process of packaging of baked products

Unit IV

- Accounts and costing :
- Book Keeping, double entry, journal, simple cash book and types of accounts.
- Purchase book, purchase return book, Stores requisition.
- Sales book, Sales return book, Cash voucher. Credit voucher book
- Preparation of invoices and debit/ credit memos
- Percentage and discounts
- Introduction to ingredient costs, labour costs, overheads, gross profit calculation of cost price, sales price and catalogue price.
- Commodities: Sugar, Cocoa, Chocolate, Milk,Butter, Cream,Cheese,Food colours, Flavors & essences
- Dry fruits & nuts used in confectionery, Fresh and preserved fruit products
- Food laws- Agmark

Practical :

1.Prepare the Bread rolls/cakes/ and assessment of its quality

2. Perform the quality test on fat/butter for bakery applications on different parameters

3. Carry out the assessment of market Candy and Chewing gum.

4. Perform the quality assessment test on yeast and skimmed milk powder for bakery application.

5. Prepare Chocolate cookies and assessment of its quality.

Reference Books

1. Samuel A. Matz, “Bakery Technology and Engineering”, Chapman & Hall, 3rd Edition, 1992.
2. Reference Books 1. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
3. Pomeranz Y, “Modern Cereal science and Technology” MVCH Publications, NY, 1987.
4. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.
5. Samuel A. Matz, “Equipment for Bakers” Pan Tech International Publication, 1988.
6. Stanley P Cauvain, Linda S Young, “Technology of Bread making”, Aspen publication,

2ndEdition, 2007.
7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.
8. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

SEMESTER II

FST 107
MANAGEMENT OF FOOD INDUSTRY

Objectives :

1. To develop a knowledge base in key areas of institutional food management.

2. To impart necessary expertise to run a food service unit.

3. To provide practical level experience in managing food service management.

4. To critically evaluate the functioning of food service units.

Unit-I

- Institutional management, Levels of Management.
- Review of different types of food service operations. Commercial, non- commercial food service institutions.
- Importance of food standards. Need for knowledge as to what constitutes good products, criteria for good quality products, hazards of poor quality, quality control, Specification for quality in food products.
- Organisation - definition and types

Unit-II

- The Manufacturing (Production) Function: Objectives of Production Management, Operation Concept, Meaning of Product Development, Factors Influencing Choice of Manufacturing Systems, Classification of Manufacturing Systems;
- Basic Procedure of Method Study (Work Study) and Time Study; Concepts, Objectives

and functions of Production Planning and Control (PPC).

- Physical plant- its location, floor plans, space allowance, Institutional food units-storage, preparation and serving

Cost control

a) Food cost, Labour cost, Maintenance cost

b) Budgets

c) Documentation and Records : Documentation and maintenance of record of raw materials and packaging materials

d) Portion control

- Role of supervisor in fruits and vegetable processing industry.
- Planning and organization of work: organization standards, process standards and procedures followed in the organization, types of products produced by the organization, Code of business conduct , Dress code.

Unit-III

- Personnel Management
- Personnel and leadership qualities for food service administration.
- Labour , Types of labour, criteria for selection and employees training.
- Labour laws and legal aspects- health & safety of employees, welfare policies.
- Manage and lead a team Team Management Skills-The Importance of Delegation, Developing Your Team, Motivating Your Team: Motivation Theories – Maslow's, Herzberg's; Worker's Participation in Management; Collective Bargaining.
- Communicating and Working With Your Team – and With Others, Communicating With People Outside Your Team
- Managing Discipline

Unit-IV

Hygiene and Sanitation

i) Personal hygiene- Importance of personal cleanliness in handling and serving food. Health examination of personnel. ii) Sanitation in handling food and equipment.

a. Preventing contamination of cooked food and handling of fresh foods,. b. Cleaning and hygienic handling of equipments. c. Insect and Rodent control d. Safety, general safety rules in food preparation and service area. Accident prevention

Equipment- Types of equipment- criteria for selection, operation and care.

Food laws and regulations on raw materials, product(s), packaging and labelling

Food safety, hygiene , sanitation GMP , HACCP and quality assurance procedures

Procedures for disposal of waste .

Books Recommended :

1. West Wood A; Harper Food Service in Institution.

2. West, Bessin, Brooks; Food Service in Institutions

3. A.M. Home Economics Association; Hand Book of Food Preparations:

4. Sweetman, M.M. 4, Mac, Keller; Food Selection and Preparation:

5. Oliver B., Watson;School Lunch Room Service

6. Lender H. Katshever and Margret E. Terrel; Food Service Planning: Layout Equipment

7. Davidson and Passmore- Human Nutrition and Dietetics.

FST 108

DAIRY TECHNOLOGY

Objectives

- To know the compositional and technological aspects of milk
- To study processed milk products.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number one is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

TECHNOLOGY OF MILK AND MILK PRODUCTS

Composition and Physical properties of milk , millk color, taste, pH and buffering capacity, refractive index, viscosity, surface tension, freezing, boiling point, specific heat, OR, electrical conductivity.

Unit II

Milk fat

- Composition and structure, factors affecting melting point, boiling point, solubility and Refractive Index, fat constants (saponification value, iodine value, RM value, Polenske value, peroxide value).
- Chemical reactions of fat (hydrolysis, auto-oxidation), condition favoring autooxidation, prevention, measurement of auto-oxidation.

Unit III

Fats and Oils

- Classification of lipids, types of fatty acids - saturated fatty acids, unsaturated fatty acids, essential fatty acids, Trans fatty acids. (Ch-3, Manay)
- Refining of oils, types- steam refining, alkali refining, bleaching, steam deodorization, hydrogenation (Ch-2 Meyer)
- Rancidity - hydrolytic and oxidative rancidity and its prevention.
- Define - margarine, butter, hydrogenated vegetable oil, lard. .

Protein and Enzymes

- General structure, amphoteric nature, difference between casein and serum protein, different types of casein (acid and rennet), uses of casein, fractionation of protein.
- Enzymes- catalase, alkaline phosphatase, lipases and proteases.

UNIT IV

- Liquid milk handling operations : Systems of collection of milk, Reception, Platform testing, Process of testing milk for accepted quality standards, organoleptic test of milk, Procedure for COB test of milk
- Various stages of processing: Filtration and Clarification,Skimming, Standardization Homogenization, Pasteurization(LTHT, HTST, UHT) Sterlization, Packaging, Storage and distribution of fluid milk
- Description and working of clarifier, cream separator, homogenizer and plate heat exchanger. Need for processing milk.
- List the various units within a dairy processing plant
- List the machineries used in a dairy processing plant
- Roles and responsibilities of a dairy products processor in a dairy processing plant

- Flow diagram of following milk products – Butter, ghee, flavored milk, yoghurt, dahi, shrikhand, ice-cream, condensed milk, milk powder, channa, paneer, cheese (cheddar)
- State the composition and nutritive value of the milk products

PRACTICAL

1. To perform platform tests in milk.(Acidity,COB,MBRT,specific gravity,SNF)
2. To estimate milk protein by Folin method.
3. To estimate milk fat by Gerber method.
4. Preparation of flavoured milk/. Pasteurization of milk
5. To prepare casein and calculate its yield.

Recommended Readings

1. De Sukumar, Outlines of Dairy Technology, Oxford University Press, Oxford. 2007
1. Hall GM, Fish Processing Technology, VCH Publishers Inc., NY, 1992
2. Sen DP, Advances in Fish Processing Technology, Allied Publishers Pvt. Limited 2005
3. Shahidi F and Botta JR, Seafoods: Chemistry, Processing, Technology and Quality, Blackie Academic & Professional,London,1994
6. Webb and Johnson, Fundamentals of Dairy Chemistry

FST 109
MILK AND DAIRY FOOD PRODUCTS DEVELOPMENT

Course Objectives

- To enable the students to understand the various dairy products and their preparation.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Unit I

- Dairy industry in India & its future prospects.
- Process of production planning
- State the factors affecting operation efficiency during production
- New Concept in milk processing- UHT, Membrane processing, Microwave and irradiation treatments; aseptic packaging. Microbiology of milk, sources of milk contamination.

Unit II

- Process for producing dairy products and process of pasteurization :
- ingredients required for production
- State the production process of pasteurization
- Process of separation and bactofugation
- Method of standardization of milk
- Method of homogenization of milk
- Method of heat exchange during pasteurization
- Process of HTST pasteurization
- Process of producing lassi, flavoured drinks, cheese dahi,kalakand, ice-cream, cream, butter , cooking butter, ghee

Unit III

- Technology of condensed and evaporated milk : Composition, nutritive value, process of manufacture, defects (their causes and prevention).
- Technology of yoghurt, Acidophilus milk, Bufgaricus milk, Kumiss, Kefir; Manufacturing of Cheddar, Mozzarella, cottage and processed cheese.
- Milk adulteration and quality control in dairy industry. By products of dairy industry and

their utilization.

- Milk based infant foods; Manufacturing of casein, Caseinate, Co-precipitates, WPC, Lactose; National and International Organizations in dairy Industry.
- Milk and milk product standards and legislations in India: Grading of milk and criterion of grading, reconstituted milk, synthetic milk.
- List the different packaging materials used to pack dairy products
- Method of packaging dairy products

Unit IV

- Methods of cleaning and sanitization: Cleaning of production area, equipment, and tools used Equipment, detergents and sanitizers used in the cleaning and maintenance of the work area, Properties of the cleaning agents used, CIP method of cleaning,
- Lubrication system and its process followed in the dairy industry
- State the different types of maintenance procedures, Periodic maintenance of all production machineries
- Method of managing and disposing waste material
- Personal hygiene and sanitation guidelines
- Demonstrate how to use tools safely
- Food safety hygiene and quality standards to follow in a work environment , HACCP principles to eliminate food safety hazards in the process and products
- Method of documenting and recording the details of raw material to final finished product

PRACTICAL :

1 Platform tests,
2 Determination of fat, SNF, TS Protein,
3 Lactose and ash contents of milk.
4 Layoput plan for setting up of milk plant,
5 Preparation and Evaluation of different types of milk and milk products 6 Preparation and evaluation of butter, ice cream, cheese, yoghurt

Text Books/References:

1 Sukumar. 1983. Outlines of Dairy Technology. De,Oxford University Press.

2. James N.Warner.1989. Principles of Dairy Processing by, Wiley Eastern Ltd.

3. Eckles,Combs; and Macy. 1982. Milk and Milk Products by,Tata McGraw Hill.

4. Aneja et al. 2002.Technology of Indian Milk Products.A Dairy India Publication.

FST 110
FOOD PACKAGING

Course Objectives

- To enable the students to understand about packaging and packaging materials, interaction of food items with packaging materials

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
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- The students are required to attempt one question from each unit.
- All questions carry equal marks

UNIT I

Introduction:

- Classify packaging, Explain the packaging process and importance of packaging.
- Roles and responsibilities of a food products packaging technician
- Equipment used for food packaging, Tools and machineries used for food packaging, Identify and set the machines and tools required for production in working condition
- Effect of environmental factors on quality of food.
- General Approach, analysis of storage requirement, accelerated storage studies: Vacuum and Inert Gas Packaging: Tests on packaging materials, Mechanical strength (Tension, notch and tearing strengths), Gas and water vapour transmission rates.
- Packaging Methods – Form fill seal packaging , Hermetic closures, retortable pouches, Aseptic packaging Inert gas packaging , Active & intelligent packaging .

Unit II

- Packaging Materials – Properties of packaging materials in relation to their function (paper, glass, jute, wood, metal containers, flexible packaging materials, laminates, tetra packs). Quality control tests on packaging materials

- Packaging Criteria – Appearance, Protection, Function and cost.
- Estimation of shelf life.
- Packaging of food products – Factors determining the packaging requirements of various foods and Description of packaging of – a. Frozen products b. Dried products c. Chemically preserved foods d. Fats and oils e. Confectionery f. Fruit juices g. Heat processed foods h. Fresh produce (Eggs, Fruits and Vegetables)
- Metal Cans As Packaging: Metallic can types - Tin cans and Aluminum cans. Specialty of Open top sanitary cans, Lacquers and their use, Three piece cans and Two piece cans, Aerosol Cans. Basics of Canning operations – Can Reformer, Flanger, Seaming, Can closures. Glass jars and Bottles in food packaging, Design features and applications, Sterilization of bottles.

Unit III

- Flexible Films Packaging: Formation of Films and pouches, Plastics used and their Specific applications.
- Copolymers their applications.
- Co-extruded films and Laminates.
- Rigid and Semi rigid plastic packaging methods.
- Extrusion –Retort pouch packaging. Laminated Paper board Cartons, Fibre Board and Corrugated Card Board packaging and their applications.

Unit IV

- Filling And Sealing Operations For Various Types Of Packages: Closing and sealing of Rigid plastic containers.
- Filling and sealing of Flexible plastic containers, Seal types, Hot wire sealing, hot bar sealing and impulse sealing. Form fill
- Seal equipment: Printing on packages, Bar codes, Nutrition labeling and legislative requirements. Filling and Sealing of pouches, pouch from fill seal machines.
- Active packaging, Moisture control, CO_2 and Oxygen scavenging, Modified atmosphere packaging –principles, applications.
- Permeability of gases in packs.
- Speciality packages.
- Personal hygiene and sanitation guidelines; • Food safety hygiene standards in a work environment, HACCP principles to eliminate food safety hazards in the process and products
- Materials and equipment used in cleaning and maintenance of work area and machinery,
- Methods of cleaning processes for tools and machineries used in packaging, Cleanliness of the process machineries required for production using recommended sanitizers
- Method of documenting and recording the details of raw material to final finished product
- Plan ,Schedule and organize the work order to prevent potential problems

PRACTICALS

1. Testing of Packaging material
2. To study the different packaging materials .
3. Determination of water vapour transmission rate for various packaging materials.
4. To determine grease resistance of packaging material.
5. To determine the wax content in given wax paper .
6. To estimate the basis weight of given packaging material.
7. To determine the chemical resistance of given packaging material.

Recommended Readings

1. Desrosier NW and Desrosier JN, The Technology of Food Preservation, CBS Publication, New Delhi, 1998
2. Paine FA and Paine HY, Handbook of Food Packaging, Thomson Press India Pvt Ltd, New Delhi- 1992
3. Potter NH, Food Science, CBS Publication, New Delhi, 1998
4. Ramaswamy H and Marcott M, Food Processing Principles and Applications CRC Press, 2006
5. Rao PG, Fundamentals of Food Engineering, PHI Learning Pvt Ltd, New Delhi, 2010
6. Toledo Romeo T, Fundamentals of Food Process Engineering, Aspen Publishers, 1999
7. Jood, Sudesh, 2002, Food Preservation , Agrotech Publisher Academy, Udaipur .
8. Potter, N.N.,2002, Food Science, CBS Publishers,
9. ND. Sethi, Mohini , 2001 Food Science, CBS Publishers,
10. ND Srilakshmi, B., 2001, Food Science, New Age International Pvt. Ltd.,
11. ND. Mahendru , S.N., 2000, Food Additives, Tata McGraw Hills, ND Manay, N.S., 2001, Foods : Facts & Principles , Wiley Eastern Ltd., ND.
12. Coles, R., Dowell, D.M., Kirwan, J. “Food Packaging Technology”. Black Well Publishing Ltd.ISBN-084939788X, 2003.
13. NIIR Board: Food Packaging Technology Handbook National Institute of Industrial Research, New Delhi (2004)
14. Pirenger O.G.and A.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH, Germany (2000).

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